Syllabus

Fall 2024, MATH1044, Applied Calculus I, Section 011

Instructor Name: Ethan Davis
Email, Phone: davis5ej@mail.uc.edu
Office Location: French Hall West, Room 4513
Office Hours (in the MASS Center, French Hall West Room 2133): Monday 1-2pm, Thursday 1-2pm, Friday 11am-12pm

The best way to contact me is through email. I will do my best to respond within 48 hours; however, if you contact me on the weekend, it may take longer.

Students are expected to review and follow the linked course policies, university policies, and community resources on the following topics in the UC Student Resources Canvas site.

- Attendance
- Class Cancellation
- Academic Integrity
- Accessibility
- Title IX (Sex-Based Discrimination/Harassment & Sexual Violence)
- Counseling Services

Who Am I?

Hi! I'm Ethan Davis, your math instructor for Fall 2024. I am a second-year grad student pursuing a PhD. I have lots of interests outside of math, so please check out my About Me page in Canvas. I'm excited to get to know all of you!

Course Format

This class is an in-person class. No in-person classes meeting will be recorded. You are responsible for what we discuss in class whether you attend or not. There is no option to complete any in-person section of MATH1044 virtually.

Course Overview, Description, Purpose

This is the first part of a two-semester sequence (MATH1044 and 1045) on calculus appropriate for students in business and life sciences. Topics covered include functions, graphs, limits, continuity, properties of exponential and logarithmic functions, differentiation, higher order derivatives, curve sketching, modeling and solving problems related to optimization and marginal analysis.

First and foremost, this course will strengthen your mathematical THINKING and problem-solving skills. We'll achieve this by introducing you to new mathematical CONCEPTS, figuring out how to solve related problems, and using the concepts to interpret information. The main concepts we will work on are functions, derivatives and their applications.

Knowing how to solve a problem isn't enough - it's equally important to be able to explain your solution to someone else. You will improve your math COMMUNICATION skills - by working with others in groups, presenting solutions in writing, and assisting your classmates.

Finally, the choice of topics and examples has been designed to give you a sense of the APPLICATIONS of calculus in your field of study. More specific learning objectives will be given to you as we progress through the course.

Pre-requisites

To take this course you must: Have taken the following courses: MATH1020 min grade C-, or MATH1021 min grade C-, or MATH1022 min grade C-, or MATH1023 min grade C-, or MATH1026 min grade C-, or MATH1059 min grade C-.

Course Learning Outcomes

University of Cincinnati e-Curriculum Outcomes:

1. Evaluate limits and apply them to difference quotients to develop the definition of a derivative.
2. Apply techniques of differentiation to polynomials, rational functions, exponential functions, logarithmic functions and trigonometric functions.
3. Use differential calculus to compute and interpret rates of change, extreme values and to sketch graphs of functions.
4. Compute higher order derivatives.
5. Apply differential calculus techniques to model, solve and interpret solutions of applied optimization problems and marginal analysis.

Ohio Transfer 36 TMM013 Business Calculus Learning Outcomes: 1.01, 1.02, 2.01, 2.02, 2.03, 3.01, 3.02, 3.03, 3.04, 3.05, 3.06, 3.07 and 4.08

Course Materials/Technology

****RECOMMENDED** Calculus Concepts**, 5th edition, by LaTorre et al.

You will also need a physical graphing calculator for this course. If you need to purchase one, please get a TI-84. If you already have a TI-83 or TI-nSpire, those are examples of other graphing calculators that are permitted. We will not allow you to use an online graphing tool during your tests.

Course Coordination/Canvas All Sections Course

MATH1044 is a large enrollment course. To make sure you have an experience that is comparable to students in other sections, a course coordinator will make sure course policies are followed, and that grading is fair across all sections. Your course coordinator is Dr. Miriam Poteet, and she can be reached by email at miriam.poteet@uc.edu for issues that cannot be resolved by your section's instructor.

Dr. Poteet communicates with all sections at the same time by setting up a Community Canvas course called **"**MATH1044 – All Sections – Fall 2024".

We need your help! In Canvas 'MATH1044 – All Sections – Fall 2024'

👍Make sure you are enrolled 📣Read all Announcements ✅Regularly check your Course Grade

Sections 011 Classroom Procedures Policies

In-Person Policy

If you are enrolled in a section that meets in-person, it is important to participate in every class. It is essential that you attend in-person for all scheduled assessments, the Midterm Exam, and the Final Exam. There is no option to complete these components remotely.

Distractions Policy

Phone and computer usage is a powerful form of distraction. Please be kind to others—silence your phone and leave the classroom to address any personal business. Only use electronic devices to be used only for course-related work during class time.

Assessments/Activities and Grading Policy

Your grade for this course is based on the following categories:

Content Checks	42% (7% each)
Midterm Exam	15%
Final Exam	20%
Homework	10%
Participation Quizzes	10%
Diagnostic Test	3%

Drop Period Ends Sept 9 @ 11:59 pm EDT (*100% Refund and class removed from academic record)*

Last day to Withdrawal Ends Nov 22 @ 11:59 pm EST (*No Refund and class remains on academic record as W)*

Course Grade/NP Grading Scheme

Letter grades will be assigned based on the percentage of the total points received. We expect to use the typical ten-point grading scale; however, at the discretion of the faculty a more generous grade scale ("curve") may be used. In other words:

- A final score of 90% will assure a grade of at least A-
- A final score of 80% will assure a grade of at least B-
- A final score of 70% will assure a grade of at least C-

Students who complete the course without earning at least a C- will be assigned the grade of NP, signifying "not proficient." An NP does not lower your grade point average; however, it also does not count as a passing grade, earn course credit or meet any pre-requisites. If this course is an essential part of your academic plan and you earn an NP (Not Proficient) as your final grade, you must retake the course successfully in a future semester.

Students who receive an NP might have earned enough points for a D+, D, or D- grade. These students can request a change in their grade from NP for up to one year after the NP grade was assigned. It is important to consider that D grades will affect your GPA and might affect your academic standing. Contact your academic advisor for help deciding whether to request this change, and/or if you have questions about the effect of the NP in your academic program.

IMPORTANT: Student must academically attend class to be eligible for an NP; this is especially important for students receiving financial aid. Attending just one class meeting does qualify as "academic attendance."

More information about academic attendance can be found here:
https://mailuc.sharepoint.com/sites/Registrar/SitePages/Class-Academic-Attendance-and-Title-IV-Compliance.aspx

To be eligible for the NP grade, a student is required to take the final exam. The final exam is a course requirement; any student that does not take the final before grades are due, will receive an "I" (or Incomplete) as their course grade until taking the final exam.

Explanation of Grading Components

Diagnostic Test This assessment is given at the end of Week 1 to gauge how prepared you are for MATH1044. If you score: ○ <63% consider joining SRS ○ <42% discuss with your instructor or academic advisor ***Re-attempts are offered during Week 2 and Week 15 only***	in-person on paper closed notes	work individually Re-attempt*
Content Checks Quizzes measure the progress you are making in mastering content. There are six (6) Content Checks this semester.	in-person on paper closed notes	work individually 1 re-attempt
Midterm Exam We will have one Midterm Exam this semester in Week 8**.**	in-person on paper closed notes	work individually no re-attempts Chapters 1, 2
Final Exam The final exam **will not** be at the day, time, and location listed on the university exam schedule, as ours is a **block exam**, scheduled by the Registrar's Office. We will be notified of the scheduled time later in the semester and will notify you as soon as possible. *Please do not make travel plans during exam week until you know the final exam date.*	in-person on paper closed notes	work individually no re-attempts, no makeups Chapters 1, 2, 3, 4
Homework The homework for this class is completed online in the All Sections Canvas Community Course. ***The average of your Midterm Exam and Final Exam will replace your homework average, if it is higher. You must attempt at least half of the assignments to be eligible for this replacement!***	online open notes	work together receive assistance unlimited re-submissions
Participation Quizzes After learning the material for each section, you will take a Participation Quiz. This will primarily be online, but I may also give these in class.	in-person online open notes	work together receive assistance unlimited re-submissions

General Assessment Policies

It is very important to us that the assessments in this class are fair, that grading is consistent, and that nobody receives unfair advantages. This is why we will strictly enforce the policies below:

- Participation in the common assessments (Content Checks, Midterm Exam and Final Exam) is mandatory.
- Cell phones and other communication devices are prohibited. They must be put away during all assessments.
- Students **must** present a photo ID and sign out with the instructor/proctor.
- If you receive accommodations, please submit your letter to your instructor as soon as possible. Registered students should schedule their assessments with Testing Services on the scheduled dates.
- If you have **documentation** of a conflict during the Midterm Exam (examples below), please submit the appropriate evidence to your instructor as soon as possible. The course coordinator will follow up with you.
    - Hospitalization/Doctor's Visit (documentation excuses you on exam date)
    - Official University Business (travel letter for athletics, clubs, internships, etc)
    - Conflict with other UC classes (submit your Catalyst schedule)
- If you believe there has been a grading error on an assessment, you can request the assessment to be re-graded within a week of your receiving it.

Final Exam

The Final Exam is scheduled by the Registrar's Office, **not** on the date listed on the Final Exam schedule. *Please do not make travel plans during exam week until you know the final exam date.* You may only request an alternate time for your Final Exam if you have a conflicting time with another course's final exam. The final exam is mandatory for all students. Any student that does not take the final before grades are due, will receive a "UW".

Re-Attempt Policies

You are only eligible for ONE re-attempt if you take a Content Check on the originally scheduled date.

Assessment	Recommended week to retake:
Content Check 1	Weeks 4, 5, 6, 15**
Content Check 2	Weeks 5, 6, 7, 15**
Content Check 3	Weeks 6, 7, 8, 15**
Content Check 4	Weeks 9, 10, 11, 15**
Content Check 5	Weeks 11, 12, 13, 15**
Content Check 6	Weeks 13, 14, 15

**There will be opportunities to complete retakes during class during Week 15.

All Retake Information will be posted in the All Sections Course.

Academic Integrity

The University Rules, including the Student Code of Conduct and other documented policies of the department, college and university related to academic integrity including the assessment policies listed above will be enforced. https://www.uc.edu/campus-life/conduct/student-code-of-conduct.html

The sanction for Academic Misconduct will be failure of the course.

(AI) Artificial Intelligence/Apps/Extensions

Generally speaking, you should not use artificial intelligence engines (e.g. ChatGPT), cell phone apps (e.g. Symbolab, Photomath) or browser extensions (e.g. Transcript, Coursology) to produce work for this class. While these might

help with checking your solutions, if misused they often provide incorrect and incomplete information. You might trick yourself into believing your calculus ability is stronger than it really is or worse—confuse yourself by introducing advanced topics beyond the scope of this course. Besides, the majority of course credit comes from in-person assessments where these resources are strictly prohibited. Use of AI on the Quizzes, Projects, Midterm Exam or Final Exam will be reported as Academic Misconduct.

The sanction for Academic Misconduct will be failure of the course.

If you are in doubt as to whether you are using an online learning support appropriately, I encourage you to meet with me for advice.

Authentic Individual Work

Although some activities of the course involve collaboration, Content Checks, the Midterm Exam and the Final Exam must be completed on your own. Violating the student code of conduct, (e.g. cheating, aiding/abetting) on these assessments, will be reported as Academic Misconduct.

The sanction for Academic Misconduct will be failure of the course.

Please make note of and review the College of Arts and Science's Two-Strikes Policy. We take these violations seriously https://www.artsci.uc.edu/information/undergrad/academic-policies.html.

Accessibility Policy

Students needing special considerations should meet with me as soon as possible to arrange reasonable provisions to ensure an equitable opportunity to meet all the requirements of this course. Some accommodations may require prior approval by the Accessibility Resources Office.

The University of Cincinnati is committed to providing all students with equal access to learning opportunities. Accessibility Resources is the official campus office that works to arrange for reasonable accommodations for students with an identified physical, psychological or cognitive disability (learning, ADD/ADHD, psychological, visual, hearing, physical, cognitive, medical condition, etc.) Students are encouraged to contact the Accessibility Resources Office to arrange for a confidential meeting to discuss services and accommodations. Contact should be initiated as soon as possible to allow adequate time for accommodations to be arranged. Additional information can be found here: https://www.uc.edu/campus-life/accessibility-resources.html

Student Religious Accommodations

Ohio law and the University's Student Religious Accommodations for Courses Policy 1.3.7 permits a student, upon request, to be absent for reasons of faith or religious or spiritual belief system or participate in organized activities conducted under the auspices of a religious denomination, church, or other religious or spiritual organization and/or to receive alternative accommodations with regard to examinations and other course requirements due to an absence permitted for the above-described reasons. Not later than fourteen days after the first day of instruction in the course, a student should provide the instructor with written notice of the specific dates for which the student requests alternative accommodations. For additional information about this policy, please contact the Executive Director of the Office of Equal Opportunity and Access at (513) 556-5503 or oeohelp@UCMAIL.UC.EDU.

MATH1044 Student Resources

Supplemental Review Sessions

All MATH1044 students are invited to sign up for an optional 12-week program of supplemental review sessions (SRS) through MATH1096. The time commitment is 1 hour and 20 minutes per week, for 12 weeks. You will review essential prerequisite material, work on problems, present solutions, and discuss and correct solutions under the guidance of upperclassmen and graduate assistants through the Learning Assistance Center. If you choose to sign

up for the SRS, then they become compulsory for you. More information about SRS can be found here: https://www.uc.edu/campus-life/learning-commons/programs/supplemental-review-sessions.html

Drop-in Study Tables

The Math and Science Support (MASS) Center provides study-table tutoring for this course. More information about the study tables can be found here: https://www.uc.edu/campus-life/learning-commons/programs/mass-center.html#study-table-schedule

Tutoring and Coaching

The Learning Assistance Center (LAC) offers one-on-one tutoring for this course. Students may schedule individual tutoring appointments to improve their understanding of course materials and develop effective study strategies. LAC appointments are available Mon-Thurs 9am-8pm and Fri 9am-5pm. Students may schedule appointments online. More information about tutoring can be found here: https://www.uc.edu/campus-life/learning-commons/programs/peer-tutoring.html

The LAC also offers Academic Coaching. Academic Coaches are high achieving UC upperclassmen and graduate students who provide one-on-one support in order to encourage success-building practices and habits in students. Coaching is not course specific, but applicable to all majors and courses. If you feel that “test anxiety'' affects your performance, consider talking to an academic coach. More information about coaching can be found here: https://www.uc.edu/campus-life/learning-commons/programs/academic-coaching.html

Communication of Changes

The professor reserves the right to make changes to the syllabus, including assignment due dates and test dates (excluding any officially scheduled examinations) when necessary to meet learning objectives or when unforeseen circumstances occur. These changes will be announced as early as possible so students can adjust their schedules. Be assured that I will communicate to you any changes to our schedule, syllabus or policies quickly and efficiently through **Canvas.** It is your responsibility to make sure Canvas has a valid email address and it is recommended that you check this email daily. All course documents will be available via link in Canvas as well.